

WealthArena

A multi-agent investing insights platform + historical fast-forward investing game

WealthArena is a production-minded platform that blends real-time market analytics, multi-instrument reinforcement-learning trading agents, explainable portfolio construction, and a historical “fast-forward” investing game. It ingests market data (stocks, ETFs, options, crypto, FX, commodities), news and alternative data, fuses signals, runs agentic trading and portfolio strategies, and exposes interactive planning, simulation and competition surfaces where users try to beat the agents or each other using historical periods.

1. Objective and End-Goal

Objective: build a modular, auditable system that generates actionable investment insights, constructs and manages risk-aware portfolios, executes or simulates trades across instruments, and offers a compelling educational/competitive game that replays historical market episodes.

End-Goal: a deployed web application + PWA/mobile-lite that allows users to: connect a profile and risk preferences, run portfolio builders and strategy labs, view explainable trade rationales and risk attributions, run backtests and paper-trades, and join historical head-to-head investment challenges. Under the hood: multi-agent RL, signal fusion, robust backtesting, and production-grade logging, experiment tracking and model serving.

2. Potential User Personas

- **Retail Investor (casual):** wants portfolio suggestions and an educational game to learn.
- **Active Trader (semi-pro):** experiments with strategies, runs simulations, compares against RL bots.

- **Quant / Student Researcher:** builds agents, tunes reward functions, needs datasets and experiment telemetry.
- **Wealth Advisor / Power User:** wants client-suitable, explainable portfolios and stress-test reports.
- **Competition Player:** uses historical-challenge mode to compete in leaderboards.
- **Business Partner / Exchange / Data Vendor:** integrates event or pricing feeds and receives analytics.

3. Value Proposition

WealthArena delivers **explainable, multi-instrument portfolio intelligence** plus a **playable historical simulation**. Users get grounded recommendations (not black-box headlines), automated risk-aware hedging suggestions, and a playground to test skill vs. learned agents. The product emphasizes auditability, reproducibility, and modularity so strategies and decisions are traceable and defensible.

4. Features

A. Market Data & Ingestion

- Multi-asset market feeds: equities, ETFs, options chains, FX, crypto, commodities.
- Macroeconomic streams: CPI, unemployment, GDP, interest rates, bond yields (FRED, ECB, OECD).
- Corporate data: earnings, filings (EDGAR), dividends, splits.
- Alternative data: news RSS, press releases, Twitter/X, Reddit, analyst sentiment, web-scraped events.
- Historical tick and minute data ingestion and downsampling pipeline.
- Data quality & monitoring (schema checks, outlier detection, Great Expectations).
- Time-series store (append-only) and canonicalization.

B. Signal Engineering & Event Extraction

- Price technical indicators (moving averages, RSI, MACD, ATR).

- Volatility estimators: realized vol, GARCH, EWMA.
- Fundamental signals: earnings surprise, revenue growth, margins, valuation ratios.
- Text-signal extraction: sentiment from news/earnings calls, named-entity (company/event), causal tagging (merger, guidance change).
- Event timeline builder: map news/events to price moves.
- Feature pipelines with versioning (DBT-like or Prefect/Airflow).

C. Agentic Multi-Agent Trading & Strategy Library

- Library of base strategies: momentum, mean-reversion, statistical arbitrage, pairs trading, sector rotation, carry strategies.
- Multi-instrument agents: stocks, ETFs, futures, options (vanilla & basic Greeks), crypto, FX.
- Strategy composition: meta-controller that composes agents by regime detection.
- Reinforcement learning agents: PPO, SAC, A2C, or off-policy (DQN variants for discrete actions).
- Hierarchical RL: high-level allocator + low-level execution policies.
- Offline RL pretraining on historical buffers; online fine-tuning in paper-trading.
- Strategy ensembling & model distillation for lightweight live execution.

D. Execution, Market Microstructure & Simulation

- Order models: market, limit, IOC, FOK; basic orderbook simulation.
- Transaction costs model: spreads, commissions, slippage, market impact (Kyle model).
- Paper-trading & live-sandbox connectors (Alpaca, Interactive Brokers sandbox, CCXT for crypto).
- Multi-asset margin & leverage simulation, position-sizing rules.
- Latency and fill simulation: configurable delay distributions and slippage.

E. Portfolio Construction & Risk Management

- Portfolio optimizers: mean-variance, Black-Litterman, Kelly fraction, risk-parity, CVaR minimization.
- Constraints: sector caps, single-name limits, liquidity, ESG filters.
- Covariance estimation: shrinkage, Ledoit-Wolf, factor models (PCA, shrinkage), dynamic covariance.

- Hedging module: delta-hedging for options, synthetic hedges using ETFs/futures.
- Scenario & stress testing: user-defined shocks (rate hikes, regime shift, sudden drawdown).
- Risk metrics: VaR, CVaR, drawdown, max drawdown, Sharpe, Sortino, turnover, exposures.
- Rebalancing policies & automated rebalancing engine.

F. Backtest, Evaluation & What-If Simulation

- Vectorized backtester with walk-forward testing.
- Bootstrapped / Monte Carlo scenario testing.
- Event-based backtests (earnings, Fed events).
- Regime-aware evaluation (separate bull/bear metrics).
- Performance attribution: factor decomposition, Brinson allocation, attribution to signals.
- Replay engine: fast-forward historical periods at configurable speed for game play.

G. News, NLP & Signal Fusion

- LLM pipeline for event extraction & reasoning (summaries of earnings calls, “tone” analyses).
- Named entity linking across filings and news.
- Cross-modal fusion: numeric + text + event embeddings into agent observation vectors.
- Alerting rules: news-driven triggers with confidence scores.
- Explainable textual rationales (model-backed) for key signals.

H. Explainability, Audit & Compliance

- Per-trade rationales: natural-language explanation + supporting evidence snippets (news, indicator states).
- Policy heatmaps & decision trace: which features drove a decision.
- Decision provenance: stored prompts, model versions, data snapshot IDs.
- Audit logs & immutable ledger of trades and model outputs for compliance.
- Model performance drift detection & alerts.

I. Game & Education Mode (Historical Fast-Forward)

- Historical challenge modes: random 3-6 month episodes get picked or artificial episodes are generated/simulated by an RL env, previous 3-12 month data provided for each episode.
- Fixed capital & instrument universe; configurable leverage/rules.
- Compete vs. agents or other users; leaderboard with performance metrics.
- Replay & rewind: step through days, inspect agent reasoning and price reactions.
- Coaching mode: annotated suggestions and “what-if” alternatives.
- Tournament manager: set theme, entry rules, and prize mechanics.
- Gamified achievements & badges for strategies / risk management.

J. User Profiles, Personalization & Portfolios

- Risk profile capture (horizon, risk aversion, liabilities).
- Goal-driven planning: target return & drawdown constraints, retirement, education goals.
- Portfolio templating & cloning: save/share portfolios and strategies.
- Adaptive suggestions: rebalance cadence and recommendations based on user behavior.

K. Analytics, Dashboards & Reporting

- Live portfolio dashboard, P&L, exposures, allocations.
- Strategy lab: run experiments, compare multiple strategies side-by-side.
- Visualization: equity curves, heatmaps, correlation matrices, option payoff diagrams.
- Exportable reports for advisory use or compliance (PDF, CSV).

L. Experimentation & Training Infrastructure

- Experiment tracking: MLflow/Weights & Biases for models, hyperparameters, backtests.
- Distributed training: Ray or Dask for parallel backtests.
- Lab for reward-model training (supervised labels from user feedback)
- **Offline Pretraining of Agents**

- Pre-train RL trading policies on historical datasets (Yahoo Finance, Alpha Vantage, Quandl, WRDS) before live deployment.
- Use **offline RL techniques** (conservative Q-learning, CQL; batch-constrained deep Q-learning, BCQ) to reduce the risk of catastrophic exploration in live markets.
- Allow simulation of different market regimes (e.g., 2008 crash, COVID shock) to stress-test agent robustness.
- **Reward Shaping & Multi-Objective Optimization**
 - Extend reward functions beyond raw returns: penalize drawdowns, volatility, overexposure to one asset class, or slippage.
 - Multi-objective reward: balance return vs. risk, liquidity, ESG goals, Sharpe ratio, max drawdown, etc.
 - Enable **customizable user-driven reward shaping**: e.g., “maximize returns but cap daily loss at 1%.”
- **Covariance-Aware Portfolio Risk Management**
 - Compute **covariance and correlation matrices** across assets dynamically.
 - Use these to construct hedges (long/short pairs, factor hedging, index vs. stock hedges).
 - Implement **risk-parity portfolio balancing** and **minimum variance optimization** (Markowitz, Black-Litterman extensions).
 - Allow agents to automatically adjust exposure in correlated instruments (e.g., hedging oil ETFs with airline shorts).

M. Ops, Monitoring & Data Governance

- Streaming & queuing: Kafka / RabbitMQ for market events, ingestion and model triggers.
- Time-series DB: TimescaleDB / ClickHouse for aggregated metrics and fast queries.
- Object store: S3 / MinIO for raw ticks, models, checkpoints.
- Caching: Redis for hot portfolios and frontend responsiveness.
- Observability: Prometheus + Grafana, Sentry for errors.
- Secrets & access control: Vault, RBAC, encryption at rest & transit.
- Data retention & reproducibility: snapshotting data versions for audits.

N. Integrations & External Connectors

- Market data: Yahoo/Alpha Vantage/Quandl/IEX/Tiingo, crypto exchange APIs.
- Broker APIs: Alpaca, Interactive Brokers, CCXT for crypto.

- Economic data: FRED, ECB, OECD.
- News & filings: RSS, EDGAR, financial news APIs.
- Payment & subscription: Stripe for premium features, billing.
- Webhooks & REST API for partners.

6. Touchpoints & Product Surfaces

- **Web Dashboard (primary):** portfolio builder, strategy lab, backtest console, live P&L, explainability panels, and game lobby.
- **Game Interface (web & PWA):** historical episode selector, play/pause/rewind controls, leaderboard, replay annotations.
- **Strategy Lab UI:** compose strategies, set reward functions, visualize agent policies (Q/value heatmaps).
- **Portfolio Builder & Allocator:** guided wizard for risk profile → portfolio suggestions → “what-if” stress tests.
- **Trade Simulator & Paper Trading Console:** manual order placement, simulated fills, and agent vs. human comparison.
- **Notifications & Alerts:** email, web push, in-app alerts for important market events or model drift.
- **Partner / Admin Portal:** data feed management, experiment dashboards, governance controls.
- **APIs & SDK:** REST/GraphQL endpoints for programmatic access to analytics, strategies, and backtest results.
- **Export & Reporting:** CSV/PDF exports, compliance-ready audit bundles.
- **Lightweight Mobile PWA:** view portfolios, receive alerts, join games.

7. Technical Architecture Recommendations & Tooling

Orchestration & Agent Coordination

- **LangGraph or LangChain Core** — multi-agent orchestration and workflow composition (discovery → signal fusion → planner → execution).
- **LangSmith** — LLM prompt/log tracking and evaluation of reasoning outputs.

Data & Storage

- **TimescaleDB / Postgres + PostGIS** — time-series & spatial queries for market data and tick metadata.
- **ClickHouse** — fast analytics for aggregated metrics and leaderboards.
- **S3 / MinIO** — raw ticks, model checkpoints, artifacts.

Ingestion & Streaming

- **Kafka** or **RabbitMQ** — streaming market feeds, ingestion, and event bus.
- **Airflow / Prefect** — scheduled ETL for daily/weekly jobs and backfill pipelines.

Models & ML Infrastructure

- **PyTorch** (primary) with optional JAX for specialized model speedups.
- **RL libraries:** Stable Baselines3, Ray-RL or RLlib, CleanRL, or custom PPO/SAC implementations.
- **Time-series & forecasting:** Temporal Fusion Transformer, N-BEATS, WaveNet/TCN.
- **Classical ML:** XGBoost / LightGBM for feature models.
- **Fine-tuning & adapters:** Axolotl or PEFT for adapter-based tuning; use for strategy LLM fine-tuning.
- **Prompt & program tuning:** DSPy (if used) for prompt/program optimization for LLM components.
- **Model serving:** BentoML, TorchServe or Triton for performant endpoints.
- **Distributed compute:** Ray for parallel backtests and distributed training.

Experimentation, Observability & Ops

- **MLflow** / Weights & Biases — experiment tracking and model lineage.
- **Prometheus + Grafana** — operational metrics.
- **Sentry** — error tracking.
- **LangSmith** — LLM logging & evaluation.
- **Optuna / Ray Tune** — hyperparameter search.

Backtesting & Simulation Tools

- Vectorized backtest engines (custom or libraries like **Backtrader**, **zipline**-style patterns).
- Monte Carlo & scenario tooling (NumPy, Pandas, custom Cython for speed).
- Orderbook / market impact simulators (custom modules or lightweight open-source libs).

Front-End & UX Stack

- **Next.js + TypeScript** — web dashboard and PWA.
- **Map & viz libs**: D3, Plotly, Deck.gl for visualizations (correlations, heatmaps, timelines).
- **State & data sync**: React Query / SWR; local caching with IndexedDB/localforage.
- **Auth & security**: OAuth2/OIDC (Keycloak/Auth0), role-based access.

CI/CD, Containerization & Deployment

- **Docker** for reproducible components; **Kubernetes** optional for scale.
- **GitHub Actions** for CI; ArgoCD for GitOps deployments.
- Secrets via Vault or cloud KMS.

Additional Tools & Utilities

- **Great Expectations** for data quality tests.
- **DuckDB / Apache Arrow / Parquet** for fast local analytics and intermediate storage.
- **Redis** for caching hot portfolios and leaderboard state.
- **Stripe** for subscriptions and payments.

Security, Compliance & Ethics

- Secure storage (KMS), RBAC, audit trails for trading decisions.
- Data minimization and synthetic-data modes for public demos.
- Clear disclaimers: not financial advice; compliance checks for markets where live trading is enabled.
- Logging & retention policies to support regulatory review.

Closing notes & pragmatic tips

- **Start small, iterate:** build core ingestion → backtest engine → simple portfolio optimizer → one RL agent → game mode. Expand instruments gradually.
- **Reproducibility first:** version datasets & models; snapshot market slices for audits.
- **Safety & legal:** get legal advice before enabling live trading; use paper trading until governance is approved.
- **Constraints:** prefer open data (Yahoo, FRED, Quandl) and free sandbox broker APIs (Alpaca paper, CCXT testnets).